MSD MANUAL
Consumer Version

Diabetes Mellitus (DM) in Children and Adolescents

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Diabetes mellitus is a disorder in which blood sugar (glucose) levels are abnormally high because the body does not produce enough insulin or fails to respond normally to the insulin produced.

- Diabetes describes a group of conditions with high blood glucose levels (hyperglycemia) caused by decreased insulin production, decreased effect of insulin, or both.
- Typical symptoms at diagnosis include excessive thirst, excessive urination, and weight loss.
- The diagnosis is based on symptoms and the results of urine and blood tests.
- Treatment depends on the type of diabetes but includes insulin injections or other medications and changes in food choices, exercise, and weight loss (if overweight).

The symptoms, diagnosis, and treatment of diabetes are similar in children and adults. However, management of diabetes in children may be more complex. It must be tailored to the child's physical and emotional maturity level and to constant variations in food intake, physical activity, and stress.

Blood sugar (glucose)

Diabetes is a disorder that affects the amount of sugar in the blood.

There are many kinds of sugar, and some kinds of sugar are a combination of two simple sugars. The white granules of sugar usually used in cooking or added to coffee or tea are sucrose. Sucrose occurs naturally in sugar cane and sugar beets. Sucrose consists of two different simple sugars, glucose and fructose. Another kind of sugar, lactose, occurs in milk. Lactose consists of the simple sugars glucose and galactose.

Sucrose and lactose must be broken down by the intestine into their simple sugars before they can be absorbed. Glucose is the main sugar the body uses for energy, so during and after absorption, most sugars are turned into glucose. Thus, when doctors talk about blood sugar, they are really talking about blood glucose.

The levels of glucose in the blood vary normally throughout the day. They rise after a meal and return to pre-meal levels within about 2 hours after eating. Once the levels of glucose in the blood return to pre-

meal levels, insulin production decreases. The variation in blood glucose levels is usually within a narrow range, about 70 to 110 milligrams per deciliter (mg/dL), or 3.9 to 6.1 millimoles per liter (mmol/L), of blood in healthy people. If people eat a large amount of carbohydrates, the levels may increase more.

Insulin

Insulin is a hormone that is released by the pancreas. Insulin controls the amount of glucose in the blood and allows glucose to move from the blood into the cells. Without a proper amount of insulin, glucose does not move into the cells and builds up in the blood. As glucose levels in the blood increase, glucose begins to appear in the urine. This glucose pulls more water into the urine, so people urinate more (polyuria) and thus become thirsty and drink more (polydipsia). Without insulin, electrolyte problems and dehydration can develop. A lack of insulin also causes fat and protein to break down.

How Insulin Works

3D MODEL

Types of Diabetes in Children and Adolescents

The types of diabetes in children are similar to those in adults. The types include

- Prediabetes
- Type 1 diabetes
- Type 2 diabetes

Prediabetes

Prediabetes is a condition in which blood glucose levels are too high to be considered normal but not high enough to be considered diabetes. Among children, prediabetes is more common among adolescents with obesity. It is temporary in over half of adolescents, but the remainder develop diabetes, especially those who continue to gain weight.

Type 1 diabetes

Type 1 diabetes occurs when the pancreas produces little or no insulin. Type 1 diabetes is the most common type among children, causing about two thirds of all cases of diabetes. It is one of the most

common chronic childhood diseases. By age 18, 1 in 300 children has developed type 1 diabetes.

Type 1 diabetes can develop at any time during childhood, even during infancy, but it usually begins between ages 4 years and 6 years or between ages 10 years and 14 years.

In type 1 diabetes, the pancreas does not produce enough insulin because the immune system attacks and destroys the cells in the pancreas that make insulin (islet cells). Such an attack may be triggered by environmental factors in people who have inherited certain genes that make them susceptible to developing diabetes. These genes are more common among certain ethnic groups (such as Scandinavians and Sardinians).

Close relatives of a person with type 1 diabetes are at increased risk of developing diabetes. Siblings have about a 6% risk, and identical twins have a more than 50% risk. The risk of diabetes for a child who has a parent with type 1 diabetes is about 4 to 9% if the father is affected and is about 1 to 4% if the mother is affected.

Children who have type 1 diabetes are at higher risk of some other disorders in which the body's immune system attacks itself (autoimmune disorders), particularly certain types of thyroid disease, and celiac disease.

Type 2 diabetes

Type 2 diabetes occurs because the cells in the body do not respond adequately to insulin (called insulin resistance). Unlike in type 1 diabetes, the pancreas can still make insulin but cannot make enough of it to overcome insulin resistance. This deficiency is often referred to as a *relative* insulin deficiency as opposed to the *absolute* deficiency seen in type 1 diabetes.

Among children, type 2 diabetes occurs mainly in adolescents but is becoming increasingly common among younger children with overweight (weigh more than 85% of children of similar age, sex, and height) or obesity (weigh more than 95% of children of similar age, sex, and height). Up until the 1990s, more than 95% of children who developed diabetes had type 1 diabetes, but now, mainly because of the increase in the number of children who have overweight or obesity, about one third of children newly diagnosed with diabetes have type 2 diabetes.

Type 2 diabetes typically develops after puberty has begun. Although many children develop type 2 diabetes between 10 years and 14 years of age, the highest rate occurs during late adolescence, between 15 years and 19 years of age (see Obesity in Adolescents). Compared to children with type 1 diabetes, children with type 2 diabetes are much more likely to have a first-degree relative (parent, sibling) or a second-degree relative (half-sibling, aunt, uncle, or grandparent) with type 2 diabetes.

Did You Know...

- Type 2 diabetes commonly occurs in people with obesity.

The increase in childhood type 2 diabetes has been particularly prominent among people who are Native American, Black, Hispanic, Asian American, and Pacific Islander.

Other children at higher risk of developing type 2 diabetes include those who

- Have high blood pressure, high blood levels of lipids (fats), obstructive sleep apnea, dark and thick skinfolds on the nape of the neck (acanthosis nigricans), fatty liver, polycystic ovary syndrome (PCOS), or a small-for-gestational-age birth weight
- Have a mother who had diabetes while pregnant (gestational diabetes) or who has a history of type 2 diabetes
- Are not physically active

Symptoms of Diabetes in Children and Adolescents

High blood glucose levels cause a variety of immediate symptoms and long-term complications.

Type 1 diabetes

Symptoms develop quickly in type 1 diabetes, usually over several days to weeks, and tend to appear in a typical pattern. High blood glucose levels cause the child to urinate excessively. Children may wet the bed or become unable to control their bladder during the day. Children who are not toilet-trained may have an increase in wet or heavy diapers. This fluid loss causes an increase in thirst and the consumption of fluids.

About half of children lose weight and have impaired growth.

Some children become dehydrated, resulting in weakness, fatigue, and a rapid pulse. Children may also have nausea and vomiting due to ketones (by-products of the breakdown of fat) in their blood. Vision may become blurred.

If the symptoms are not recognized as being caused by diabetes and treated, children may develop a life-threatening disorder called diabetic ketoacidosis.

Type 2 diabetes

Many children do not have any symptoms or have only mild symptoms, and their type 2 diabetes may be detected only when blood or urine tests are done for other reasons (such as during a physical before playing sports or going to camp).

Symptoms in children with type 2 diabetes are milder than those in type 1 diabetes and develop more slowly. Parents may notice an increase in the child's thirst and urination or only vague symptoms, such as fatigue.

Children with type 2 diabetes are less likely to develop ketoacidosis than those with type 1 diabetes, but ketoacidosis or a type of severe dehydration and confusion (called hyperosmolar hyperglycemic state) can still develop.

Complications of Diabetes in Children and Adolescents

Diabetes can cause immediate complications and long-term complications. The most serious immediate complication is diabetic ketoacidosis.

Long-term complications are usually due to mental health issues or to blood vessel problems. Although blood vessel problems take years to develop, the better the control of diabetes, the less likely that complications will occur.

Diabetic ketoacidosis (DKA)

DKA is present at the time of diagnosis in up to one fourth of children with type 1 diabetes, and sometimes is present at the time of diagnosis in children with type 2 diabetes.

DKA is also common among children with previously diagnosed type 1 diabetes. It develops in about 1 to 10% of children with type 1 diabetes each year, usually because these children have not taken their insulin. DKA also can develop in children who have had previous episodes of it, are facing difficult social circumstances, or are depressed or having other mental health issues, which may impact how they manage their diabetes. Problems with insulin delivery (for example, problems with their insulin pump) can rapidly lead to DKA. DKA can also occur if children do not receive enough insulin during illness (when ill, children need more insulin).

Without insulin, cells cannot use the glucose that is in the blood. Cells switch to a back-up mechanism to obtain energy and break down fat, producing compounds called ketones as by-products.

Ketones make the blood too acidic (ketoacidosis), causing nausea, vomiting, fatigue, and abdominal pain. The ketones make the child's breath smell like nail polish remover. Breathing becomes deep and rapid as the body attempts to correct the blood's acidity (see Overview of Acid-Base Balance). Some children develop a headache and may become confused or less alert. These symptoms may be caused by accumulation of fluid in the brain (cerebral edema).

DKA, when untreated, can progress to coma and death. Children with DKA are also dehydrated and often have other chemical imbalances in the blood, such as abnormal levels of potassium and sodium.

Mental health issues

Mental health issues (see Support) are common among children with diabetes. Up to half of children develop depression, anxiety, or other psychological issues (see Overview of Mental Health Disorders Children and Adolescents).

Because insulin can cause weight gain, eating disorders are a serious problem in adolescents, who sometimes skip their insulin doses to try to control their weight.

Mental health problems can affect children's ability to follow their meal plan and medication regimens, which means their blood glucose is poorly controlled.

Blood vessel effects

Diabetes eventually causes narrowing of small and large blood vessels. The narrowing can damage many different organs. Although the blood vessel narrowing starts to develop within a few years after diabetes begins, the organ damage usually does not become apparent until years later and is rarely present during childhood.

Damage to small blood vessels most often affects the eyes, kidneys, and the nerves. Damage to the blood vessels of the eyes as a result of diabetes (called diabetic retinopathy) can cause loss of vision. Damage to the kidneys (called diabetic nephropathy) can result in kidney failure. Damage to the nerves (called diabetic neuropathy) can result in numbness, tingling, or burning pain in the arms and legs. These problems are more common among children who have type 2 diabetes than type 1 diabetes. These problems also may be present at the time of diagnosis or earlier in children who have type 2 diabetes.

Damage to large blood vessels most often involves the arteries to the heart and the brain. Changes to blood vessels in children with diabetes may contribute to high blood pressure. Narrowing of arteries to the heart can cause heart attack. Narrowing of arteries to the brain can cause stroke. Heart attack and stroke do not commonly occur during childhood but can occur later in life.

Diagnosis of Diabetes in Children and Adolescents

- Blood glucose tests
- Hemoglobin A1c (HbA1C) test
- Sometimes an oral glucose tolerance test
- Determination of diabetes type (type 1, type 2)

The diagnosis of diabetes is a two-part process. Doctors first determine whether children have diabetes and then determine the type. Children who appear to have complications also have other testing.

Diagnosing diabetes in children

Doctors suspect diabetes when children have typical symptoms or when a urine test done during a routine physical examination reveals glucose. To confirm the diagnosis, doctors measure blood glucose levels.

Blood glucose levels can be measured in the morning before children eat (called the fasting glucose level) or without regard to meals (called the random glucose level). Children are considered to have diabetes if they have both typical symptoms of diabetes *and* a high blood glucose level. If the fasting glucose level is 126 mg/dL (7.0 mmol/L) or higher on 2 tests done at different times, children have diabetes. If the random glucose level is 200 mg/dL (11.1 mmol/L) or higher, children probably have diabetes but should have their fasting glucose level tested to confirm.

Doctors also measure the level of a protein in the blood called hemoglobin A1c (HbA1C). Hemoglobin is the red, oxygen-carrying substance within red blood cells. When blood is exposed to high blood glucose

levels over a period of time, glucose attaches to the hemoglobin and forms HbA1C. Because HbA1C takes a relatively long time to form and to break down, levels change only over weeks to months rather than from minute to minute like blood glucose levels do. HbA1C levels thus reflect blood glucose levels over a 2- to 3-month period of time. Children whose HbA1C level is 6.5% or higher are considered to have diabetes. HbA1C levels are especially helpful for diagnosing type 2 diabetes in children who do not have typical symptoms.

Another kind of blood test called an oral glucose tolerance test may be done in children who have no symptoms or whose symptoms are mild or not typical. In this test, children fast, have a blood sample taken to determine the fasting glucose level, and then drink a special solution containing a large amount of glucose. Doctors then measure blood glucose levels 2 hours later. If the level is 200 mg/dL (11.1 mmol/L) or higher, children are considered to have diabetes. This test is similar to the test that pregnant women have to look for gestational diabetes.

LAB TEST

Hemoglobin A1C (HbA1c) Test

Diagnosing the type and stage of diabetes

To help distinguish type 1 diabetes from type 2, doctors do blood tests that detect antibodies to various proteins made by the insulin-producing cells in the pancreas. Antibodies are important to fight off foreign substances such as germs, but sometimes antibodies attack normal cells. In the case of diabetes, cells that make insulin and other chemicals related to insulin are examples of normal cells that can be attacked. Such antibodies are usually present in children with type 1 diabetes and are rarely present in children with type 2 diabetes. Type 1 diabetes is an example of an autoimmune disorder.

After type 1 diabetes is diagnosed, doctors can determine the stage. Type 1 diabetes progresses in stages:

- Stage 1: Children have 2 or more diabetes-specific antibodies (antibodies are measured by blood tests) but have normal blood glucose levels and no symptoms of diabetes.
- Stage 2: Children have 2 or more diabetes-specific antibodies, abnormal blood glucose levels, and usually no symptoms of diabetes.
- Stage 3: Children have 2 or more diabetes-specific antibodies, high blood glucose levels, and symptoms of diabetes.
- Stage 4: Children have type 1 diabetes with severe symptoms (such as kidney malfunction with protein in the urine).

Testing after diagnosis

Children who are diagnosed with type 1 diabetes usually have other blood tests to look for other autoimmune disorders such as celiac disease and thyroid disease. These tests are done at diagnosis and every 1 to 2 years thereafter.

Doctors sometimes do tests to look for other problems such as a disorder of the adrenal glands (Addison disease), joint and muscle disorders (such as juvenile idiopathic arthritis), and additional digestive tract disorders (such as inflammatory bowel disease).

Children who are diagnosed with type 2 diabetes have blood tests to determine how their liver and kidneys are functioning and urine tests. At diagnosis, children who have type 2 diabetes are also tested for other problems, such as high blood pressure, high blood levels of lipids (fats), and fatty liver, because these problems are common among children with type 2 diabetes. Other tests are done depending on symptoms. For example, children with daytime sleepiness or who snore while sleeping are tested for obstructive sleep apnea, and adolescent girls who are hairy and have acne or menstrual irregularities are tested for polycystic ovary syndrome.

Treatment of Diabetes in Children and Adolescents

- Nutrition and exercise
- For type 1 diabetes, injections of insulin
- For type 2 diabetes, metformin and sometimes insulin or other medications

The main goal of diabetes treatment is to keep blood glucose levels as close to the normal range as can be done safely. However, no treatment completely maintains blood glucose at normal levels. When people try very hard to keep blood glucose levels normal, they increase the risk that their blood glucose levels will sometimes become too low. Low blood glucose is called hypoglycemia and can be dangerous.

Although advances in diabetes technology have improved quality of care and control of blood glucose, not all people have benefited. In the United States, children who are White or non-Hispanic have a lower rate of complications and poor outcomes. Race, ethnicity, socioeconomic status, neighborhood and physical environment, access to healthy foods, and access to health care are examples of other factors that contribute to whether a child with diabetes can successfully control their blood glucose.

Children with diabetes should carry or wear medical identification (such as a bracelet or tag) to alert emergency care professionals to the presence of diabetes. This information allows health care professionals to start life-saving treatment quickly, especially in the case of injury or change in mental status.

Nutrition and exercise

Children with either type of diabetes need to

- Make healthy food choices
- Lose weight if overweight
- Exercise regularly

General nutritional management and education are particularly important for all children with diabetes. Dietary recommendations for children with diabetes are based on healthy eating

recommendations for all children and aim to maintain ideal body weight and optimal growth and to prevent short-term and long-term complications of diabetes.

All children should eat regularly and not skip meals. Although most dietary regimens allow some flexibility in carbohydrate intake and meal times, having meals and scheduled snacks at about the same time each day and that contain similar amounts of carbohydrates is important for optimal glucose control. Because carbohydrates in food are turned into glucose by the body, variations in carbohydrate intake cause variations in blood glucose levels.

Choosing healthy foods can help control blood glucose and protect heart health. Children should focus on eating fruits and vegetables, whole grains, and high-fiber foods (for example, foods that have at least 3 grams of fiber or more per serving). Food should not contain many highly processed (refined) carbohydrates, particularly candy, baked goods (such as cookies, donuts, and pastries), and sugary drinks. Children should have no more than 4 to 8 ounces of 100% fruit juice per day. They should avoid regular soda, sweetened iced tea, lemonade, fruit punch, and sports drinks altogether. Children also should avoid foods with saturated fats, such as baked goods, snack foods (such as potato chips and corn tortilla chips), deep-fried foods (such as french fries), and fast food. Some of these foods may still contain trans fats, common ingredients in certain commercial foods, which are being removed because they have been shown to be associated with an increased risk of heart disease.

In **type 1 diabetes,** parents and older children are taught how to gauge the carbohydrate content of food and to develop a meal plan. In most children with type 1 diabetes, food intake is not rigidly specified and is based on the child's usual eating patterns and insulin doses are matched to actual carbohydrate intake. Infants and preschool-aged children present a particular challenge to parents because they do not eat consistent amounts of food and because they may develop hypoglycemia but may not be able to communicate symptoms of hypoglycemia to their parents.

In **type 2 diabetes,** lifestyle modifications focus on weight in the majority of children. Steps to improve food choices and manage food intake include eliminating sugary drinks, controlling portion size, switching to low-fat foods, and increasing fiber by eating more fruits and vegetables.

Regular exercise is important because it improves glucose control and makes it easier to lose weight. Because vigorous exercise can cause a significant drop in blood glucose, some children with type 1 diabetes may need to consume some extra carbohydrates before and/or during a workout.

Monitoring blood glucose

The frequency of monitoring depends on the type of diabetes.

In **type 1 diabetes,** blood glucose levels should be checked up to 6 to 10 times per day and should be measured before all meals, before a bedtime snack, during illness, and if children have symptoms of low blood glucose (hypoglycemia) or high blood glucose (hyperglycemia). Self-monitoring with a fingerstick glucose test is most often used to monitor blood glucose. Most blood glucose–monitoring devices (glucose meters) use a drop of blood obtained by pricking a fingertip (fingerstick) with a small implement called a lancet. The lancet holds a tiny needle that can be jabbed into the finger or placed in a spring-loaded device that easily and quickly pierces the skin. The drop is placed on a test strip and the

test strip is read by a machine (glucometer). The machine reports the result on a digital display. Because exercise can lower glucose levels for up to 24 hours, glucose should be measured more frequently on days children exercise or are more active. Sometimes levels need to be measured during the night.

In **type 2 diabetes,** blood glucose levels should be measured regularly but typically less often than in type 1 diabetes. Several factors determine the frequency of self-monitoring, including children's glucose levels between meals and after eating. The frequency of monitoring should increase to at least 3 times a day if children do not have good control of their glucose, during illness, or when symptoms of hypoglycemia or hyperglycemia are felt. Once glucose is controlled, home testing is limited to a few between-meal and after-meal blood glucose measurements per week.

Once experience is gained, parents and many children can adjust the insulin dose as needed to achieve the best control. In general, by 10 years of age, children start to become interested in testing their own blood glucose levels and injecting insulin themselves. Parents should encourage this independence but make sure the child is being responsible. Doctors teach most children how to adjust their insulin dosage in accordance with the patterns of their home blood glucose records.

Parents should use a journal, app, spreadsheet, smart meter, or cloud-based program to keep detailed daily records of all factors that can affect control of blood glucose, including blood glucose levels, timing and amount of insulin doses, carbohydrate intake, physical activity, and any other relevant factors (for example, illness, late snack, or a missed insulin dose).

Children with either type of diabetes typically see their doctor several times a year. The doctor evaluates their growth and development, reviews blood glucose records that a parent keeps or that are captured by a monitoring device, provides guidance and counseling about nutrition, and measures glycosylated hemoglobin levels (hemoglobin A1C). The doctor usually screens for long-term complications of diabetes by measuring protein in the urine, doing tests to determine how the thyroid gland is functioning (thyroid function tests), doing tests to look for nerve damage, and doing eye examinations. Screening tests may be done once a year or at other intervals.

Continuous glucose monitoring (CGM) systems are an increasingly common method of monitoring blood glucose levels and can replace routine self-monitoring of blood glucose for some children. In CGM systems, a small glucose sensor placed under the skin measures blood glucose levels every 1 to 5 minutes, 24 hours a day. They transmit real-time results of blood glucose levels wirelessly to a device that may be built into an insulin pump, to a wireless monitor that can be worn on a belt, or to a smartphone or smartwatch app. The systems also record results for the doctor to review. Alarms on CGM systems can be set to sound when blood glucose levels drop too low or climb too high, so the devices can help people with type 1 diabetes quickly identify worrisome changes in blood glucose that they can treat right away. Use of CGM devices may help lower HbA1C levels.

Two types of CGM systems are currently available: real-time CGM and intermittently scanned CGM.

Real-time CGM can be used in children 2 years of age and older. This type of system automatically transmits a continuous stream of glucose data to the user in real time, provides alerts and active alarms, and also transmits glucose data to a receiver, smartwatch, or smartphone. Real-time CGM should be done as close to daily as possible for maximum benefit.

Intermittently scanned CGM can be used in children 4 years of age and older. This type of system provides the same type of glucose data as real-time CGM but requires the user to purposely scan the sensor to obtain information. Newer intermittently scanned CGM systems have optional alerts and alarms. Intermittently scanned CGM should be done frequently, at least once every 8 hours. Children who use a CGM device need to be able to measure blood glucose with a fingerstick to calibrate their monitor and to verify glucose readings if they do not match their symptoms.

Although CGM devices can be used with any insulin regimen, they are typically worn by insulin pump users. When used in conjunction with an insulin pump, the combination is known as **sensor-augmented pump therapy**. This therapy requires manual adjustment of insulin doses based on CGM data results.

Other CGM systems are integrated with a pump and can also lower the insulin dose if blood glucose drops too low. This integration can reduce the number of episodes where blood glucose drops too low, even when compared to sensor-augmented pump therapy.

Closed-loop insulin pumps can be used in children 2 years of age and older. They automatically provide the right amount of insulin by using sophisticated computer algorithms that are on a smartphone or similar device and link a CGM sensor to an insulin pump to determine blood glucose levels and control insulin delivery. Current closed-loop systems are not truly automated because they require users to manually provide insulin for meals and snacks and make adjustments for exercise. These systems help to more tightly control insulin dosing and limit episodes where blood levels of insulin are too high or too low. A fully automated closed-loop system, sometimes known as an artificial pancreas, continues to be evaluated and is not yet commercially available.

Did You Know...

- Children with type 1 diabetes always need insulin injections, regardless of their weight or what they eat.

Type 1 diabetes treatment

To control blood glucose, children with type 1 diabetes take injections of insulin.

When type 1 diabetes is first diagnosed, children are usually hospitalized. Children with type 1 diabetes are given fluids (to treat dehydration) and insulin. They always require insulin because nothing else is effective. Children who do not have diabetic ketoacidosis (DKA) at diagnosis typically receive 2 or more daily injections of insulin. Insulin treatment is usually begun in the hospital so that blood glucose levels can be tested often and doctors can change insulin dosage in response.

After diagnosis, children must take insulin regularly. Doctors work with children and their family to determine which insulin regimen is best.

There are several types of insulin regimens:

- Multiple daily injections (MDI) regimen using a basal-bolus regimen
- Insulin pump therapy
- Fixed forms of MDI regimen or premixed insulin regimen (less common)

Most children who have type 1 diabetes should be treated with MDI regimens or with insulin pump therapy.

A **basal-bolus regimen** is the preferred MDI regimen. This regimen involves taking one injection of a longer-acting insulin (basal dose) every day and then separate supplemental injections (bolus doses) of a short-acting insulin immediately before meals. Each bolus dose can be different depending on how much food the child is going to eat or what the blood glucose level is at that time.

An advantage of the basal-bolus regimen is that it allows for flexibility as to when meals are eaten and how much is eaten.

In **insulin pump therapy,** the basal dose of insulin is delivered through a small, flexible tube (catheter) that is left in the skin. Supplemental boluses that are given at mealtime or that are given to correct a high blood glucose level are given as separate injections of rapid-acting insulin via the insulin pump.

Insulin pump therapy is increasingly being used in children. Potential benefits include better glucose control, safety, and user satisfaction compared to MDI regimens. This therapy is preferred for younger children, such as toddlers and preschoolers, and overall offers an added degree of control to many children.

Fixed forms of MDI regimens are less commonly used. If a basal-bolus regimen is not an option (for example, if adequate supervision is not available, such as when an adult is not available to give injections at school or daycare), fixed forms of MDI regimens can be an option. In these regimens, children typically receive a specific (fixed) amount of shorter-acting insulin before eating breakfast and dinner and a fixed dose of longer-acting insulin at bedtime.

Fixed regimens provide less flexibility, require a daily set schedule for meals, and have been largely replaced by basal-bolus regimens wherever possible.

Premixed insulin regimens use a fixed mixture of two forms of insulin: one that works quickly and lasts for only a few hours, and one that takes longer to work but lasts longer. The usual ratios of insulin are 70/30 (70% longer-acting and 30% shorter-acting) or 75/25. Children are given one injection at breakfast and one at dinner.

An advantage of premixed regimens is that they require fewer injections and are easier to manage. However, premixed regimens have less flexibility with respect to timing and amount of meals and cannot be adjusted as frequently. Thus, these regimens do not control blood glucose levels as well as other regimens.

Insulin delivery methods

Insulin can be injected in several ways:

- Vial and syringe
- Insulin pen
- Insulin pump

Some children use a **vial and syringe**. In this method, each dose of insulin is drawn up into a syringe from a vial and is injected under the skin, usually in the arm, thigh, or abdominal wall. Small syringes with very thin needles make the injections nearly painless. The amount of insulin the syringe holds varies depending on the amount of insulin needed per injection. Young children often use syringes with 1/2 unit markings to allow for smaller adjustments in the doses of insulin to be used.

An **insulin pen** is a convenient way for many children to carry and use insulin, especially for children who take several injections a day outside the home. The pen contains a cartridge that holds enough insulin for several doses. The dose delivered on each injection is adjusted by turning the top of the pen.

Another device is an **insulin pump,** which automatically pumps insulin continuously from a reservoir through a catheter that is left in the skin. The catheter site must be changed every 2 to 3 days. More and more children, even young children, are using insulin pumps. The pump more closely mimics the way the body normally delivers insulin. Pumps are programmed to release small doses of insulin continuously over 24 hours (called the basal dose) and can be triggered manually to deliver extra insulin (called the bolus dose) with meals or to treat high blood glucose. Unlike other methods, insulin pumps use only short-acting insulin. Children do not need longer-acting insulin because they are continually receiving a small amount of insulin in the basal dose. The pump can be programmed to give different amounts of insulin at different times of day and night.

Insulin pumps may be used with continuous glucose monitoring systems (see Continuous glucose monitoring (CGM) systems) to better track trends in blood glucose throughout the day. Newer insulin pumps have been developed that combine insulin pump therapy with continuous glucose monitoring systems in one device.

For some children, the pump offers an added degree of control, whereas others find wearing the pump inconvenient or develop sores or infections at the catheter site. Children must rotate their injection and pump sites to avoid developing lipohypertrophy. Lipohypertrophy is an accumulation of fatty lumps of tissue under the skin. The lumps occur at injection sites that have been overused for insulin and can cause blood glucose levels to vary because they can prevent insulin from being absorbed consistently.

Type 2 diabetes treatment

Children with type 2 diabetes usually are not treated in the hospital unless the diabetes is severe. Usually they are given medications to lower blood glucose levels (antihyperglycemic medications) at a regular doctor's office visit. Children with severe diabetes may need to be hospitalized to start insulin treatment. Less commonly, children with type 2 diabetes develop severe dehydration or, as in type 1 diabetes, diabetic ketoacidosis (DKA).

Metformin is most commonly the first medication given by mouth (orally) for children and adolescents under 18 years of age. It is started at a low dose and often increased over several weeks to higher

doses. It can be taken with food or given as an extended-release formulation to prevent nausea and abdominal pain.

Insulin is given to children who are hospitalized because of ketosis, DKA, or hyperosmolar hyperglycemic state. Insulin can often be stopped after several weeks once glucose levels return to normal after treatment with metformin. Children whose type 2 diabetes is not controlled by metformin alone are given insulin or another medication called liraglutide. About half of adolescents with type 2 diabetes ultimately require insulin.

Liraglutide, exenatide, and **dulaglutide** are injectable medications that can be given to children over 10 years of age who have type 2 diabetes. **Semaglutide** is another injectable medication that can be given to people 12 years of age and over to manage type 2 diabetes and to treat obesity. These medications are known as GLP-1 agonists. GLP-1 is a hormone that has several roles in the body, including preventing more glucose from going into the bloodstream, slowing stomach emptying, and affecting the areas of the brain that process hunger and fullness (satiety). The GLP-1 agonists work by acting like the GLP-1 hormone and therefore help manage blood glucose by triggering the pancreas to release more insulin and by having an effect that reduces appetite and hunger, resulting in weight loss. GLP-1 agonists also help lower HbA1C levels. They may be given to children who are taking metformin but whose HbA1C level is not in the target range, or they can be given instead of metformin to children who cannot tolerate that medication.

Empagliflozin is a medication taken by mouth that can be given to children over 10 years of age with type 2 diabetes. This medication is a sodium-glucose cotransporter-2 (SGLT2) inhibitor. Empagliflozin helps lower blood glucose levels by increasing the amount of glucose that is removed from the body in urine. It cannot be taken by people who have severe kidney disease or who are on dialysis. It can increase the risk of DKA and may cause urinary tract infections (UTIs) and genital yeast infections.

Other medications used for adults with type 2 diabetes may help some adolescents, but they are more expensive, and there is limited evidence for their use in children.

Some children who lose weight, improve their food choices, and exercise regularly may be able to stop taking the medications.

Treatment of complications of diabetes

Diabetic ketoacidosis (DKA) is usually treated in an intensive care unit. They often require fluids given by vein (intravenously) to correct dehydration. They often also need intravenous potassium solutions to correct low potassium levels. Children often require intravenous insulin during DKA.

To prevent the development of DKA and minimize the need for hospitalization, children and families should use ketone test strips to check for ketones in blood or urine. Blood testing may be preferred in younger children and in others in whom it is difficult to obtain a urine sample, those who have frequent episodes of DKA, and insulin pump users. Ketone testing should be done whenever children become ill (regardless of the blood glucose level) or when the blood glucose is high. High ketone levels may indicate DKA, especially if children also have abdominal pain, vomiting, drowsiness, or rapid breathing.

Hypoglycemia is low blood glucose that occurs when too much insulin or too much of an antihyperglycemic medication is taken or when the child does not eat regularly or exercises vigorously for a long period of time. Warning symptoms include confusion or other abnormal behavior, and children often appear pale and/or sweaty.

To treat hypoglycemia, children are given sugar in any form, such as glucose tablets, hard candies, glucose gel, or a sweet drink, such as a glass of fruit juice. If children are unable to eat or drink (for example, because they are confused, disoriented, are having a seizure, or are unconscious), an injection of glucagon is given.

If untreated, severe hypoglycemia causes weakness, confusion, and even coma or death.

In adults, adolescents, and older children, episodes of hypoglycemia rarely cause long-term problems. However, frequent episodes of hypoglycemia in children younger than 5 years of age may impair intellectual development. Also, young children may not be aware of the warning symptoms of hypoglycemia. To minimize the possibility of hypoglycemia, doctors and parents monitor young children with diabetes particularly closely and allow a slightly higher target range for their blood glucose level. Continuous glucose monitoring systems can help children because these systems sound an alarm when glucose falls below a specified range.

Adolescents with diabetes

Some children with diabetes do very well and control their diabetes without undue effort or conflict. In others, diabetes becomes a constant source of stress within the family, and control of the condition deteriorates. Adolescents may have particular problems controlling their blood glucose levels because of

- Hormonal changes during puberty: These changes affect how the body responds to insulin. As a result, higher doses are usually needed during this time.
- Adolescent lifestyle: Peer pressure, increased activities, erratic schedules, concern about body image, or eating disorders may interfere with the prescribed treatment regimen, particularly their meal plan.
- Experimentation with alcohol, cigarettes, and illicit drugs: Adolescents who experiment with these substances may neglect their treatment regimen and may be at greater risk of complications of diabetes (such as hypoglycemia and DKA).
- Conflicts with parents and other authority figures: Such conflicts may make adolescents less willing to follow their treatment regimen.

Thus, some adolescents need a parent or another adult to recognize these issues and give them the opportunity to discuss problems with a health care professional. A health care professional can help make sure adolescents remain appropriately focused on keeping their blood glucose levels under control. Parents and health care professionals should encourage adolescents to check their blood glucose levels frequently.

Adolescents benefit if the doctor considers their desired schedule and activities and takes a flexible approach to problem solving by working with the adolescent rather than imposing solutions.

Support

Mental health problems affect children with diabetes and their families. The realization that they have a lifelong condition may cause some children to become sad or angry, and sometimes even deny that they have an illness. Parents can look for a doctor, psychologist, or counselor who can address these emotions and help the child adhere to the required regimen of meal plan, physical activity, blood glucose testing, and medications. Untreated mental health problems can lead to difficulties controlling blood glucose.

Summer camps for children with diabetes allow these children to share their experiences with one another while learning how to become more personally responsible for their condition.

For the treatment of diabetes, the child's primary care doctor usually enlists the aid of a team of other professionals, possibly including a pediatric endocrinologist, dietitian, diabetes educator, social worker, or psychologist. Family support groups may also help. The doctor may provide parents with information to bring to school so that school personnel understand their roles.

Screening and Prevention of Diabetes in Children and Adolescents

Type 1 diabetes

Given the high rate of progression to stages of type 1 diabetes where people have symptoms and the prolonged period before symptoms are present, therapies have been studied in an effort to prevent or delay the start of type 1 diabetes (stage 3).

A monoclonal antibody, teplizumab, can delay the start of type 1 diabetes in people 8 years of age and older who do not yet have symptoms of diabetes (stage 2). People are given an infusion of teplizumab once a day for 14 days. It can cause fever, nausea, fatigue, and headaches in the first 5 days and can also cause a decrease in white blood cells called lymphocytes (lymphopenia). Teplizumab can delay the start of symptoms of type 1 diabetes for about 2 years.

Type 2 diabetes

Because prompt measures (such as changes in food choices, an increase in physical activity, and weight loss) may help prevent or delay the start of type 2 diabetes, children at risk of type 2 diabetes should be screened with a blood test that measures the hemoglobin A1C levels. This test should first be done when children are 10 years old or when puberty starts (if puberty occurred at a younger age) and should be repeated every 3 years if normal.

Some risk factors for type 2 diabetes can be prevented. For example, children who are obese should lose weight, and all children should get regular exercise (see Nutrition and exercise).

More Information

The following English-language resources may be useful. Please note that THE MANUAL is not responsible for the content of the resources.

American Diabetes Association: Comprehensive information on diabetes, including resources for living with diabetes

JDRF (previously called Juvenile Diabetes Research Foundation): General information on type 1 diabetes

National Institute of Diabetes and Digestive and Kidney Diseases: General information on diabetes, including on the latest research and community outreach programs

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